

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 23

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Unburnt fuel in vehicle exhaust gases (1) • Evaporation from fuel tanks / spillage at filling stations (1) • Solvents (paints, cleaning products) (1) • Aerosol propellants (1) • Gas leaks / fossil fuel exploitation (methane) (1)	3
01.2	<i>1 mark for each point.</i> • Vapours displaced from the vehicle tank are collected, condensed and returned to the main storage tank (1) • Vapours are passed through an activated carbon filter where the hydrocarbons adsorb onto the carbon (1)	2
01.3	<i>1 mark for each point.</i> • A platinum / palladium catalyst on a high surface-area honeycomb in the exhaust pipe (1) • Carbon monoxide and hydrocarbons are oxidised to carbon dioxide (and water) (1) • NO_x are chemically reduced to nitrogen and oxygen (1)	3
01.4	<i>1 mark for each point.</i> • Binds to haemoglobin (more strongly than oxygen) so haemoglobin cannot carry oxygen from the lungs to tissues (1) • Brain damage / death if enough haemoglobin is inactivated; low doses worsen chronic heart disease (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 249, 254; AQA spec 3.4.3.2.4-3.4.3.2.5.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> • B (about 40:40:20 sand:silt:clay) (1) • Balanced mix gives good drainage and aeration from the sand fraction (1) • Water retention and nutrient (ion) retention from the clay fraction; easy root penetration and cultivation (1)	3
02.2	<i>1 mark for each point.</i> • Large sand particles leave large pore spaces so water drains rapidly (high permeability / aeration) (1) • Nutrient ions adsorb onto the (charged) surfaces of clay particles but not onto sand (1) • With only 5% clay there is little surface to hold ions so nutrients are leached; low water retention too (1)	3
02.3	<i>1 mark for each point.</i> • C (clay) holds much water so has a high thermal capacity: warms and cools slowly; A warms rapidly after cold weather (1) • Clay particles held together by capillary water so roots find it hard to push between them; penetration is easy in sandy A (1)	2
02.4	<i>1 mark for each point.</i> • Shake a known mass of soil with water in a measuring cylinder and leave to settle; particles settle in order of size (sand first, then silt, then clay; organic matter floats) (1) • Measure the depth / volume of each layer to calculate the percentage of each fraction, then use the soil triangle to name the textural class (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-160, 412; AQA spec 3.2.5.1.	
03.1	<i>1 mark for each point.</i> • The underlying granite contains uranium whose decay produces the radioactive gas radon (an alpha emitter) (1) • Radon travels up cracks in the rock and through foundations into buildings (1) • It becomes concentrated inside enclosed / poorly ventilated buildings; control by ventilating foundations / sumps (1)	3
03.2	<i>1 mark for the correct answer.</i> • $16 + 1.5 = 17.5\%$ (1)	1
03.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Where they live (geology / radon, altitude) (1) • Occupation (radiographer, nuclear worker, aircrew, miner) (1) • Medical history (number of X-rays / CT scans) (1) • Lifestyle / diet (eg shellfish near discharges); smoking (1)	2
03.4	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • X-rays carry a small risk of mutation / cancer but the benefit (diagnosis, avoiding invasive surgery) may outweigh it; risks can be reduced (shielding, dose limits) (1) • Symptoms of chronic low-dose exposure take a long time to develop (1) • The same symptoms (cancers) have other causes so effects are hard to attribute (1) • Accurate data on human exposure / doses is often unavailable (especially after accidents) (1) • People who benefit and people who take the risk may be different (eg uranium miners vs electricity consumers) (1)	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 297-299; AQA spec 3.4.3.2.14.	
04.1	<i>1 mark for each point.</i> - The scatter / variability of results around each mean; about 68% of values lie within one SD of the mean (1) - The SDs are large relative to the difference in means (1.9 g) so the ranges overlap greatly (1)	2
04.2	<i>1 mark for each point.</i> - t-test (1) - Comparing two means of measured (continuous) data (1)	2
04.3	<i>1 mark for each point.</i> - There is a 30% probability that the difference occurred by random chance (1) - Above the 0.05 (5%) level so the difference is not significant; the null hypothesis (no difference) is accepted (1)	2
04.4	<i>1 mark for each point.</i> - Two correlated variables may both be caused by a third variable that was not investigated (1) - Significance only shows the result is unlikely to be chance; a mechanism / controlled experiment is needed for causation (1)	2
04.5	<i>1 mark for each point.</i> - Increase the number of replicates (reduces the effect of variability) (1) - Standardise other variables (soil, water, light) more closely to reduce scatter (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
05.1	<i>1 mark for each point.</i> - Cost increases as the percentage prevented increases (1) - The relationship is not linear: cost rises slowly at first then very steeply / exponentially above about 80-90% (1)	2
05.2	<i>1 mark for the correct answer.</i> $16.0 - 5.0 = 11.0$ (£ million) (1)	1
05.3	<i>1 mark for each point.</i> - ALARA: emissions should be As Low As Reasonably Achievable (1) - It is rarely practical / affordable to reduce emissions to zero; low-level emissions may have acceptable impacts so an acceptable emission level is set (1) - BATNEEC: Best Available Technology Not Entailing Excessive Cost - choosing the most effective control equipment whose cost is not excessive (1) - The factory would install equipment that gives a high level of control (eg 80-90%) rather than paying £16 million for 99% (1)	4

Question	Marking guidance	Marks
05.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Industry / city may have expanded so there are more pollutant sources and total emissions have risen (1) • Better understanding of the health / environmental effects so stricter standards are set (1) • Improved technology becomes available (BATNEEC changes) / cost of better control falls (1) • Pollutant may have accumulated (persistent) so background levels are higher (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244-245; AQA spec 3.4.3.1.	
06.1	<i>Correct answer only.</i> • C (1)	1
06.2	<i>1 mark for each point.</i> • Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms (1) • Tropospheric ozone is a secondary pollutant (from NO_x photochemistry) that is toxic to plants and humans and is a greenhouse gas (1)	2
06.3	<i>1 mark for each point.</i> • HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere (1) • So most do not persist long enough to reach the stratosphere where UV would release their chlorine (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	
07.1	<i>1 mark for each point.</i> • No large dam / reservoir needed so no flooding of farmland, homes or habitats and no barrier across the whole river (run-of-river / micro-hydro diverts part of the flow) (1) • Suitable for rivers with a small drop in height and lower flow rates that are common in lowland rural areas (1) • Small scale matches local demand; electricity used locally without long transmission lines / grid connection (1) • Low construction cost and simple technology maintained by the community (1) • Helical turbines tolerate high turbidity / debris without damage and fish pass through unharmed; Kaplan turbines with adjustable blades cope with variable flow (1)	5
07.2	<i>1 mark.</i> • Lack of capital / finance, technical knowledge or grid access; seasonal drying of rivers; conflicts with irrigation water; planning / land ownership (1)	1
07.3	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Output is proportional to flow x efficiency: $100 \times 0.10 \times (40 / 90)$ (1) • = 4.4 kW (accept 4.4-4.5) (1)	2

Question	Marking guidance	Marks
07.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Constant regulated flow removes seasonal extremes: dry-season sandbanks used by nesting birds / turtles lost; high flows that clear gravel spawning beds lost; floodplain no longer flooded (1) • Sudden flow increases for peak generation raise turbidity, wash away species, reduce light for photosynthesis (1) • Sediment trapped so less nutrient replenishment downstream; barrier to migrating fish (salmon, sturgeon) and isolation of populations (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 203-205; AQA spec 3.3.4.1.	
08.1	<i>1 mark for each point.</i> • Most energy comes from non-renewable fossil fuels whose reserves are depleting; the rate of new discoveries has fallen sharply since the 1980s (1) • Use causes unacceptable environmental damage: atmospheric pollution, climate change, oil pollution, habitat loss (1) • Economic / social: depleting resources become more expensive, reducing future affluence; risk of an energy gap if new technologies are not developed in time (1)	3
08.2	<i>1 mark for each point.</i> • Coal mining: habitat loss (especially open-cast), spoil heaps, subsidence from deep mines, acid drainage, dust, noise (1) • Oil and gas extraction: spills from rigs, oil-based drilling mud polluting groundwater and sea, gas flaring releasing SO₂ / CO₂ / smoke (1) • Pipelines causing habitat damage during construction; tanker transport risks (1) • Fuel processing (refineries, coal washing) pollution; embodied energy of equipment and site development for all resources (1)	4
08.3	<i>1 mark for each point.</i> • New technologies take many years to develop, prove and scale up, and need new infrastructure (grids, storage, vehicles) (1) • Existing cheap fossil fuel technologies make it hard for alternatives to compete until fuel prices rise (1) • By then depletion would reduce supply faster than replacements could be built, causing shortages (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 180-186; AQA spec 3.3.3.	
09.1	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • High persistence means DDT remains in the environment for years / decades, so it has time to disperse and be taken up (1) • High liposolubility means it dissolves in fat and is stored in body tissues rather than excreted (bioaccumulation) (1) • Concentration increases at each trophic level as predators eat many contaminated prey (biomagnification) (1) • Birds of prey are at the top of the food chain so receive the highest concentrations (1) • Sub-lethal doses caused thin eggshells that broke / reduced breeding success; higher doses caused sterility or death (1)	4

Question	Marking guidance	Marks
09.2	<i>1 mark for each point.</i> • Advantage: low persistence / low liposolubility so they do not bioaccumulate or biomagnify / break down quickly (1) • Disadvantage: high mammalian toxicity so farm workers / humans are at risk of acute poisoning (neurotoxin inhibiting acetylcholinesterase); chronic effects (memory loss, depression); suspected carcinogens (1)	2
09.3	<i>1 mark.</i> • Pyrethroids are highly toxic to fish (even though mammal toxicity is low) (1)	1
09.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Neonicotinoids are water soluble so they are absorbed by the roots (from the soil) and translocated through the plant (systemic) (1) • They are moderately persistent so remain in the soil long enough to be absorbed (1) • Seed coating avoids spraying so no spray drift / no dust onto flowers or surrounding habitats (1) • So bees / pollinators are less exposed than when the crop is sprayed (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-265; AQA spec 3.4.3.2.8.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Carbon: combustion of fossil fuels and biomass moving carbon from long-term stores to the atmosphere; deforestation reducing photosynthesis; ploughing / drainage increasing aerobic decomposition; new anaerobic environments (paddies, landfill, reservoirs, livestock) releasing methane; marine pollution reducing phytoplankton; consequences: rising CO₂ and CH₄, climate change, ocean acidification, reduced soil organic matter • Nitrogen: Haber process doubling fixation; legume cultivation; drainage increasing nitrification and reducing denitrification; combustion NO_x; sewage and fertiliser leaching; consequences: eutrophication, smogs, acid rain, N₂O warming, NO_x toxicity • Counter-strategies carbon: renewables / nuclear replacing fossil fuels, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, increasing soil organic matter (low tillage, mulching), CCS (pre / post combustion, storage) • Counter-strategies nitrogen: reduced combustion, catalytic converters / urea sprays, biological fixation instead of Haber, legumes and crop rotation, management of biological wastes as fertiliser, low-solubility fertilisers, timing, buffer strips, low tillage, nitrate vulnerable zones • Evaluation: some strategies well established and cheap (rotation, buffer strips), others expensive or unproven (CCS); scale of fossil fuel dependence; feedback and time lags; link to circular economy and low-temperature natural processes; judgement	9
10.2	<i>1 mark.</i> • The average time an atom / molecule remains in the reservoir before being transferred out (volume / transfer rate) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-157; AQA spec 3.2.4.	

Question	Marking guidance	Marks
11.1	Discuss the strategies that can be used to provide sustainable supplies of water for a growing population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Context: uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • New sources: rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • Management: artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • Conservation: low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering; pollution control to protect sources • Treatment: matching quality to use; screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	25

Question	Marking guidance	Marks
11.2	Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Attributes: accuracy, precision, representative samples, reliability, anomalies, validity; hypothesis and null hypothesis; preliminary studies • Sampling design: random vs systematic location and avoiding bias; timing (diurnal, seasonal, long-term, weather); sample size and number and the running mean; standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • Statistics: variability, standard deviation and error bars, significance and p-values, correlation vs causation; choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • Application to Paper 1 topics: monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	25